Measuring a Nation's Income

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IN THIS CHAPTER

- What is Gross Domestic Product (GDP)?
- How is GDP related to a nation's total income and spending?
- What are the components of GDP?
- How is GDP corrected for inflation?
- Does GDP measure society's well-being?

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Economics

- Microeconomics
  - Study of how households and firms
    - Make decisions
    - Interact in markets
- Macroeconomics
  - Study of economy-wide phenomena
    - Including inflation, unemployment, and economic growth

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Income and Expenditure

- Gross Domestic Product (***GDP***)
  - Measures total income of everyone in the economy.
  - Also measures total expenditure on the economy's output of goods and services.
- Income equals expenditure
  - For the economy as a whole
  - Because every dollar a buyer spends is a dollar of income for the seller.

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Active Learning 1: Income Equals Expenditure

- Nalah pays James $50 to mow her lawn.
  - A. What happens with total expenditure?
  - B. What happens with total income?
- James is a seller of a service and Nalah is a buyer: James earns $50, and Nalah spends $50.
  - A. Total expenditure rises by $50.
  - B. Total income rises by $50.

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The Circular-Flow

- The Circular-Flow Diagram
  - Simple depiction of the macroeconomy
  - Illustrates ***GDP*** as spending, revenue, factor payments, and income
- Preliminaries:
  - Factors of production: inputs like labor, land, capital, and natural resources.
  - Factor payments: payments to the factors of production (e.g., wages, rent).

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The Circular-Flow Diagram – 1

Households:

- own the factors of production, sell or rent them to firms for income
- buy and consume goods & services

Firms

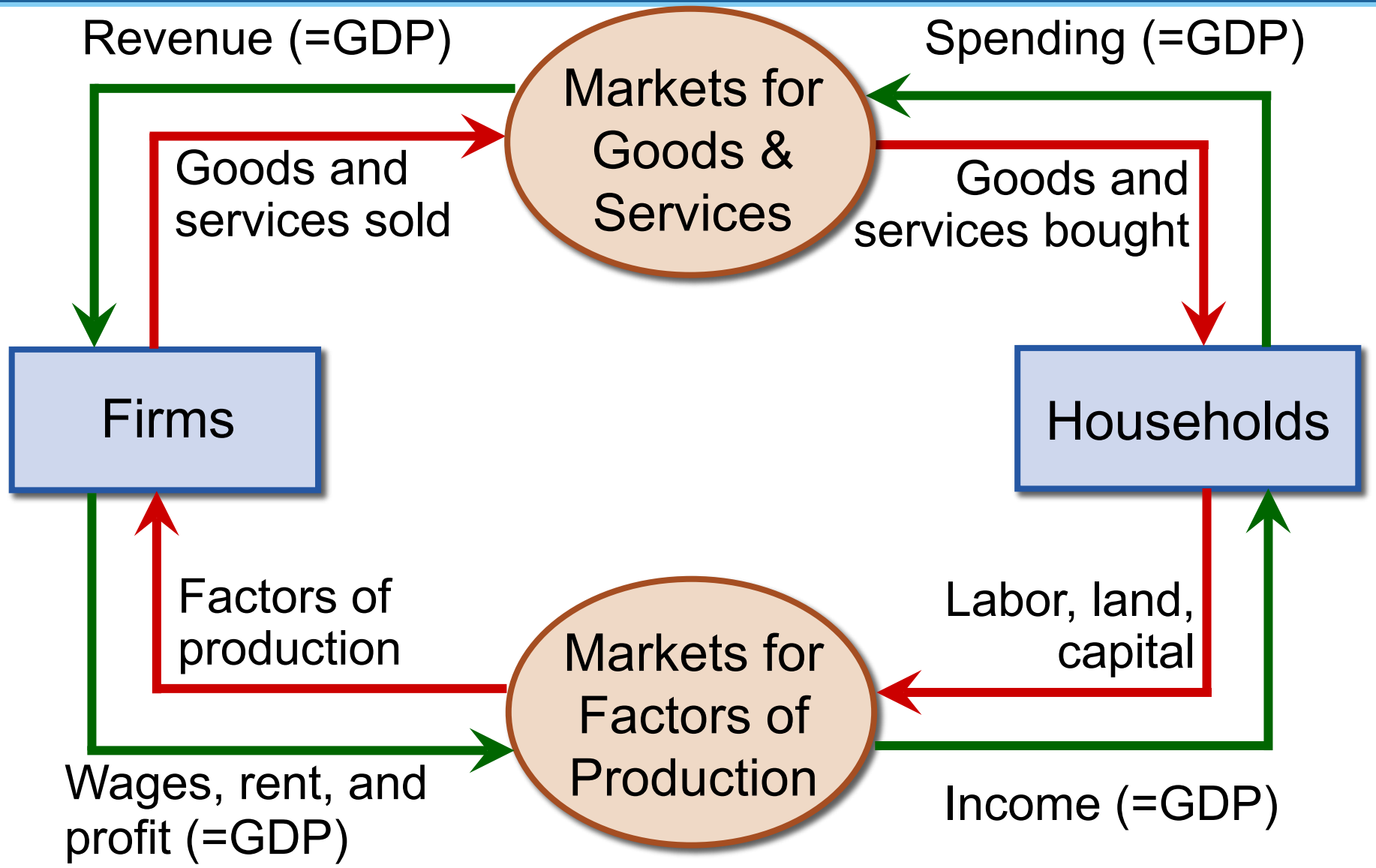
Households

Firms:

- Buy or hire factors of production, use them to produce goods and services
- sell goods & services

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The Circular-Flow Diagram – 2



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What This Diagram Omits

- The government
  - Collects taxes, buys goods and services
- The financial system
  - Matches savers' supply of funds with borrowers' demand for loans
- The foreign sector
  - Trades goods and services, financial assets, and currencies with the country's residents

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Gross Domestic Product (***GDP***) Is… – 1

- …the **market value** of all final goods & services produced within a country in a given period.
  - Goods are valued at their market prices, so:
    - All goods measured in the same units (e.g., dollars in the U.S.)
    - Things that don't have a market value are excluded, e.g., housework you do for yourself.

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Gross Domestic Product (***GDP***) Is … – 2

- …the market value of **all** final goods & services produced within a country in a given period.
  - ***GDP*** includes all items produced in the economy and sold legally in markets
  - ***GDP*** excludes most items produced and sold illicitly. It also excludes most items that are produced and consumed at home.

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Gross Domestic Product (*GDP*) Is … – 3

- …the market value of all **final** goods & services produced within a country in a given period.
  - Final goods: intended for the end user
  - Intermediate goods: used as components or ingredients in the production of other goods
- ***GDP*** only includes final goods
  - They already embody the value of the intermediate goods used in their production.

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Gross Domestic Product (*GDP*) Is … – 4

- …the market value of all final **goods & services** produced within a country in a given period.

  - ***GDP*** includes tangible goods (like food, mountain bikes, beer)
  - and intangible services (dry cleaning, concerts, haircuts).

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Gross Domestic Product (***GDP***) Is … – 5

- …the market value of all final goods & services **produced** within a country in a given period.

  - ***GDP*** includes currently produced goods, not goods produced in the past.

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Gross Domestic Product (*GDP*) Is … – 6

- …the market value of all final goods & services produced **within a country** in a given period.

  – ***GDP*** measures the value of production that occurs within a country's borders, whether done by its own citizens or by foreigners located there.

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Gross Domestic Product (***GDP***) Is … – 7

- …the market value of all final goods & services produced within a country in a **given period**.

  - ***GDP*** measures the value of production that takes place within a specific interval of time
  - Usually a year or a quarter (3 months)

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Active Learning 2: What is included in ***GDP***?

Nalah pays James $50 to mow her lawn.

A. What happens with GDP?

B. Will your answer to the previous question change if Nalah and James get married?

A. When Nalah pays James $50 to mow her lawn, that transaction is part of GDP.

B. If James and Nalah get married, and James continues to mow Nalah's lawn, the value of the mowing is now left out of GDP because James' service is no longer sold in a market.

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The Components of *GDP*

- Recall: ***GDP*** is total spending.
- Four components:
  - Consumption (***C***)
  - Investment (***I***)
  - Government Purchases (***G***)
  - Net Exports (***NX***)
- These components add up to ***GDP*** (denoted ***Y***): $Y = C + I + G + NX$

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Consumption (*C*)

- Consumption, ***C***
  - Spending by households on goods and services
  - Excludes: purchases of new housing
- Note on housing costs:
  - For renters, ***C*** includes rent payments.
  - For homeowners, ***C*** includes the imputed rental value of the house, but not the purchase price or mortgage payments

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Investment (*I*)

- Investment, ***I***
  - Spending on goods that will be used in the future to produce more goods
    - Business capital: business structures, equipment, and intellectual property products
    - Residential capital: landlord's apartment building; a homeowner's personal residence
    - Inventory accumulations: goods produced but not yet sold

"Investment" does not mean the purchase of financial assets like stocks and bonds.

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Government Purchases (*G*)

- Government purchases (*G*)
  - Spending on the goods and services purchased by the government
    - At the centre, state, and local levels.
  - Excludes transfer payments
    - Such as Social Security or unemployment insurance benefits.
    - They are not purchases of goods and services

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Net Exports (*NX*)

- Net exports, ***NX*** = exports – imports
  - Exports: foreign spending on the economy's goods and services
  - Imports: are the portions of ***C***, ***I***, and ***G*** that are spent on goods and services produced abroad

- Adding up all the components of ***GDP*** gives:

$$Y = C + I + G + NX$$

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U.S. *GDP* and its Components, 2021

	Billions	*% of GDP*	*Per person*
Y	$22,996	100.0	$69,132
C	15,742	68.5	47,323
I	4,120	17.9	12,386
G	4,053	17.6	12,184
NX	-918	-4.0	-2,760

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Active Learning 3: ***GDP*** and its components

What is the effect on ***GDP*** and its components?

A. Jahzara spends $300 to buy her husband dinner at the finest restaurant in Boston.

B. Nylah spends $1,200 on a new smartphone to use in her publishing business. The smartphone was built in China.

C. Joseph spends $800 on a tablet to use in his editing business. He got last year's model on sale for a great price from a local manufacturer.

D. GM builds $500 million worth of cars, but consumers only buy $470 million worth of them.

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Active Learning 3: Answers A, B

A. Jahzara spends $300 to buy her husband dinner at the finest restaurant in Boston.

- ***C*** rises by $300
- ***GDP*** rises by $300.

B. Nylah spends $1,200 on a new smartphone to use in her publishing business. The smartphone was built in China.

- ***I*** rises by $1,200
- ***NX*** fall by $1,200 (because Imports rise by $1,200)
- ***GDP*** is unchanged.

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Active Learning 3: Answers C, D

C. Joseph spends $800 on a tablet to use in his editing business. He got last year's model on sale for a great price from a local manufacturer.

- ***I*** and ***GDP*** are unchanged
  - ***I*** (Joseph's business) rises by $800
  - ***I*** (inventories) fall by $800

D. GM builds $500 million worth of cars, but consumers only buy $470 million of them.

- ***C*** rises by $470 million
- ***I*** (inventory) rises by $30 million
- ***GDP*** rises by $500 million.

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Real versus Nominal *GDP*

- Nominal ***GDP***
  - The production of goods and services valued at current prices
  - Not corrected for inflation
- Real ***GDP***
  - The production of goods and services valued at constant prices (base year)
  - Is corrected for inflation
- For the base year

  Nominal ***GDP*** = Real ***GDP***

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EXAMPLE 1A: Calculating Nominal *GDP*

Compute total spending (nominal ***GDP***) in each year:

	Pizza		Chocolate	
year	***P***	***Q***	***P***	***Q***
2021	$15	300	$2.50	1,000
2022	$16	400	$3.00	1,200
2023	$17	500	$3.50	1,300

2021: $15 x 300 + $2.50 x 1,000 = $7,000

2022: $16 x 400 + $3.00 x 1,200 = $10,000

2023: $17 x 500 + $3.50 x 1,300 = $13,050

Increase:

42.9%

30.5%

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EXAMPLE 1B: Calculating Real *GDP*

Compute real ***GDP*** in each year (base year 2021):

	Pizza	$15	Chocolate	$2.50
year	***P***	***Q***	***P***	***Q***
2021	**$15**	300	**$2.50**	1,000
2022	$16	400	$3.00	1,200
2023	$17	500	$3.50	1,300

Increase:

2021: $15 x 300 + $2.50 x 1,000 = $7,000

28.6%

2022: $15 x 400 + $2.50 x 1,200 = $9,000

19.4%

2023: $15 x 500 + $2.50 x 1,300 = $10,750

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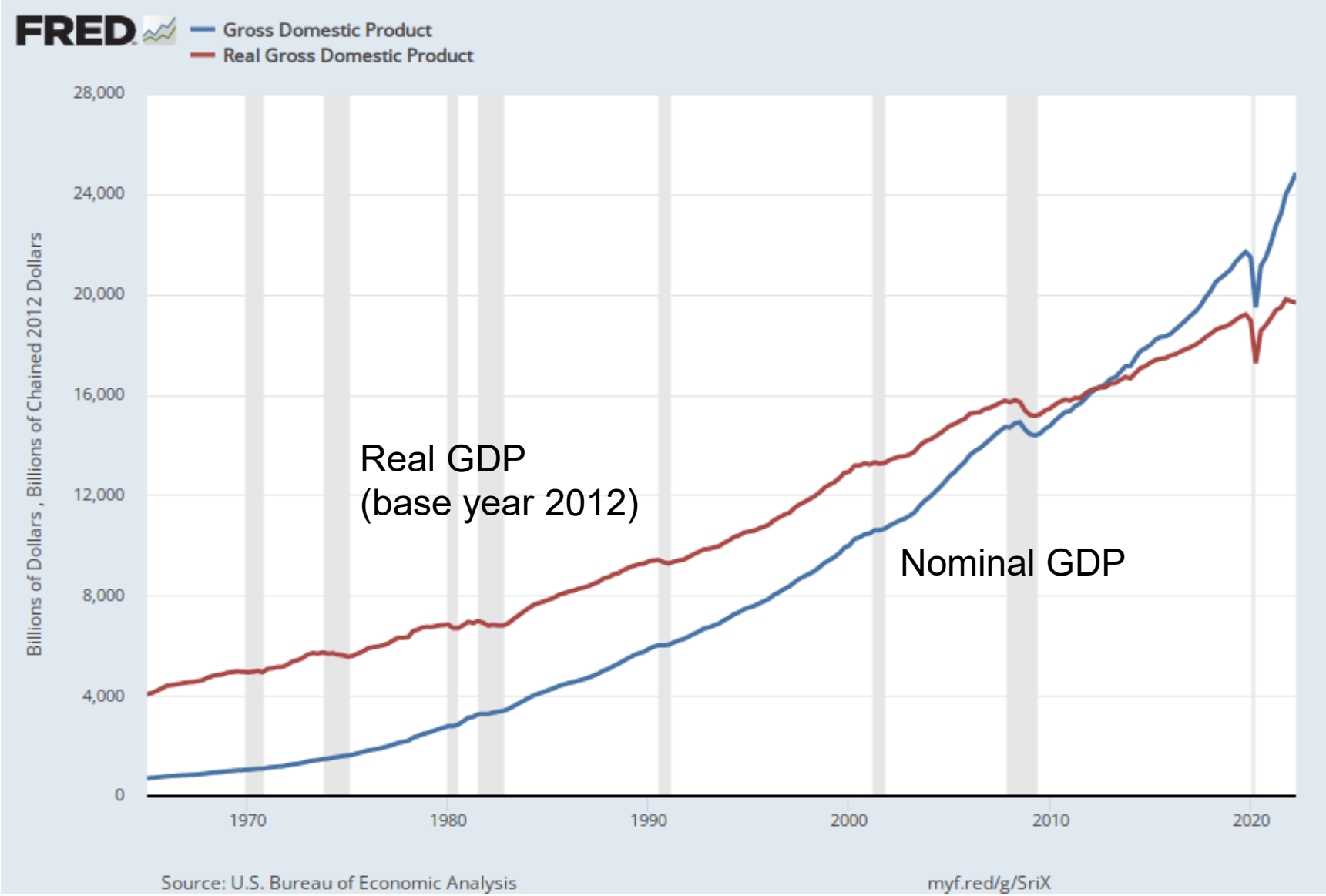
EXAMPLE 1C: Correcting for Inflation

year	***Nominal GDP***	***Increase:***	***Real GDP***	***Increase:***
2021	$7,000	**42.9%**	$7,000	**28.6%**
2022	$10,000		$9,000	
2023	$13,050	**30.5%**	$10,750	**19.4%**

- The change in nominal ***GDP*** reflects both changes in prices and quantities.
- The change in real ***GDP*** is the amount that ***GDP*** would change if prices were constant (i.e., if zero inflation) – reflects only changes in quantities.
- ***Hence, real GDP is corrected for inflation.***

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Nominal and Real GDP in the U.S., 1965–2022



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The *GDP* Deflator

- ***GDP*** deflator $= 100 \times \frac{Nominal\ GDP}{Real\ GDP}$
  - A measure of the price level
  - Measures the current level of prices relative to the level of prices in the base year
- Economy's inflation rate
  - Compute the percentage increase in the GDP deflator from one year to the next

$$\text{Inflation rate in year 2} = \frac{\text{GDP deflator in year 2} - \text{GDP deflator in year 1}}{\text{GDP deflator in year 1}} \times 100.$$

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EXAMPLE 1D: Calculate the *GDP* Deflator

Compute the *GDP* deflator in each year:

year	***Nominal GDP***	***Real GDP***	***GDP deflator***	***Inflation rate***
2021	$7,000	$7,000	**100.0**	
2022	$10,000	$9,000	**111.1**	**11.1%**
2023	$13,050	$10,750	**121.4**	**9.3%**

2021: 100 x (7,000/7,000) = 100

2022: 100 x (10,000/9,000) = 111.1

2023: 100 x (13,050/10,750) = 121.4

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Active Learning 4: Computing *GDP*

	2021 (base year)		2022		2023	
	P	Q	*P*	Q	*P*	Q
Cookies	$1	900	$2	1,000	$3	1,250
Smartphones	$900	185	$1,000	200	$1,200	210

Use the above data to solve these problems:

A. Compute nominal ***GDP*** in 2021.

B. Compute real ***GDP*** in 2022.

C. Compute the ***GDP*** deflator in 2023.

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Active Learning 4: Answers A, B

	2021 (base year)		2022		2023	
	P	Q	*P*	Q	*P*	Q
Cookies	$1	900	$2	1,000	$3	1,250
Smartphones	$900	185	$1,000	200	$1,200	210

A. Compute nominal ***GDP*** in 2021.

$1 x 900 + $900 x 185 = $167,400

B. Compute real ***GDP*** in 2022.

$1 x 1,000 + $900 x 200 = $181,000

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Active Learning 4: Answers, C

	2021 (base year)		2022		2023	
	P	Q	*P*	Q	*P*	Q
Cookies	$1	900	$2	1,000	$3	1,250
Smartphones	$900	185	$1,000	200	$1,200	210

C. Compute the ***GDP*** deflator in 2023.

Nom ***GDP*** = $3 x 1,250 + $1,200 x 210 = $255,750

Real ***GDP*** = $1 x 1,250 + $900 x 210 = $190,250

GDP deflator = 100 x (Nom ***GDP***)/(Real ***GDP***)

= 100 x ($255,750)/($190,250) = 134.4

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GDP and Economic Well-Being

- Real ***GDP*** per capita
  - Main indicator of the average person's standard of living

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GDP Does Not Measure:

- The quality of the environment
- Leisure time
- Non-market activity
  - Child-care a parent provides at home
  - Volunteer work
- An equitable distribution of income

- ***GDP*** is a good measure of economic well-being for most—but not all—purposes.

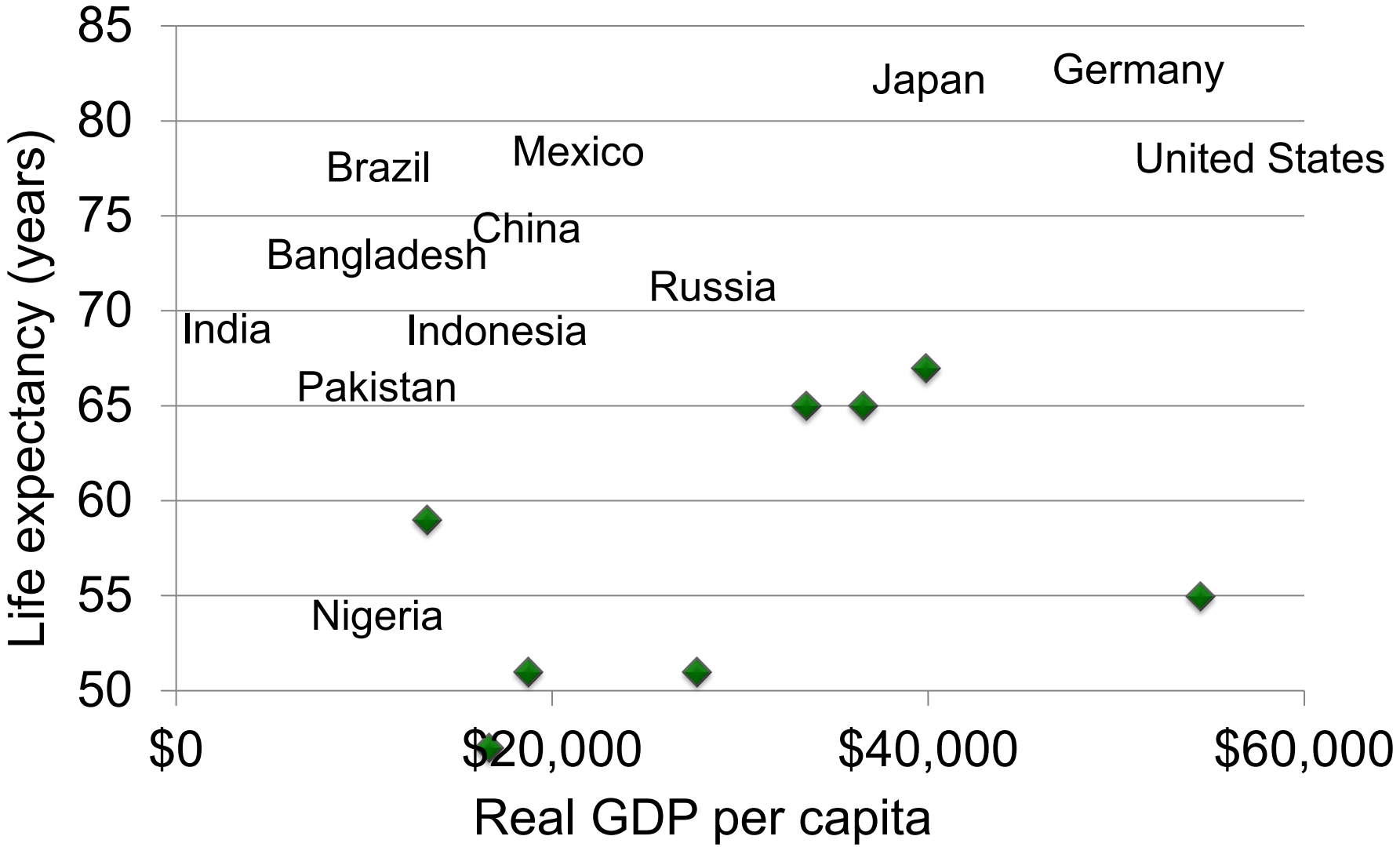
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Then Why Do We Care About ***GDP***?

- Having a large ***GDP*** enables a country to afford
  - Better schools, a cleaner environment, health care, etc.
- Many indicators of the quality of life are positively correlated with ***GDP***. For example…

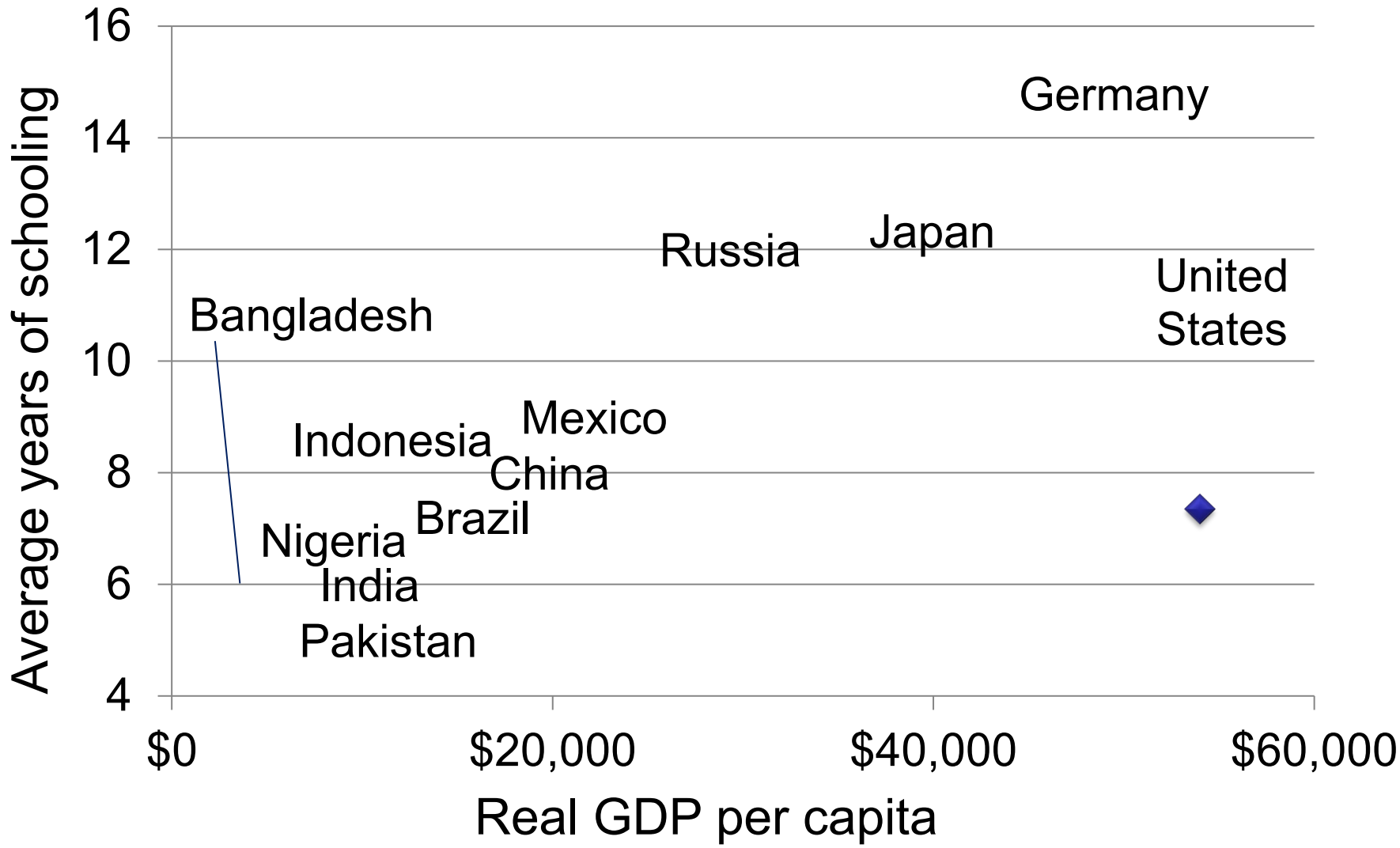
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GDP and Life Expectancy in 12 Countries



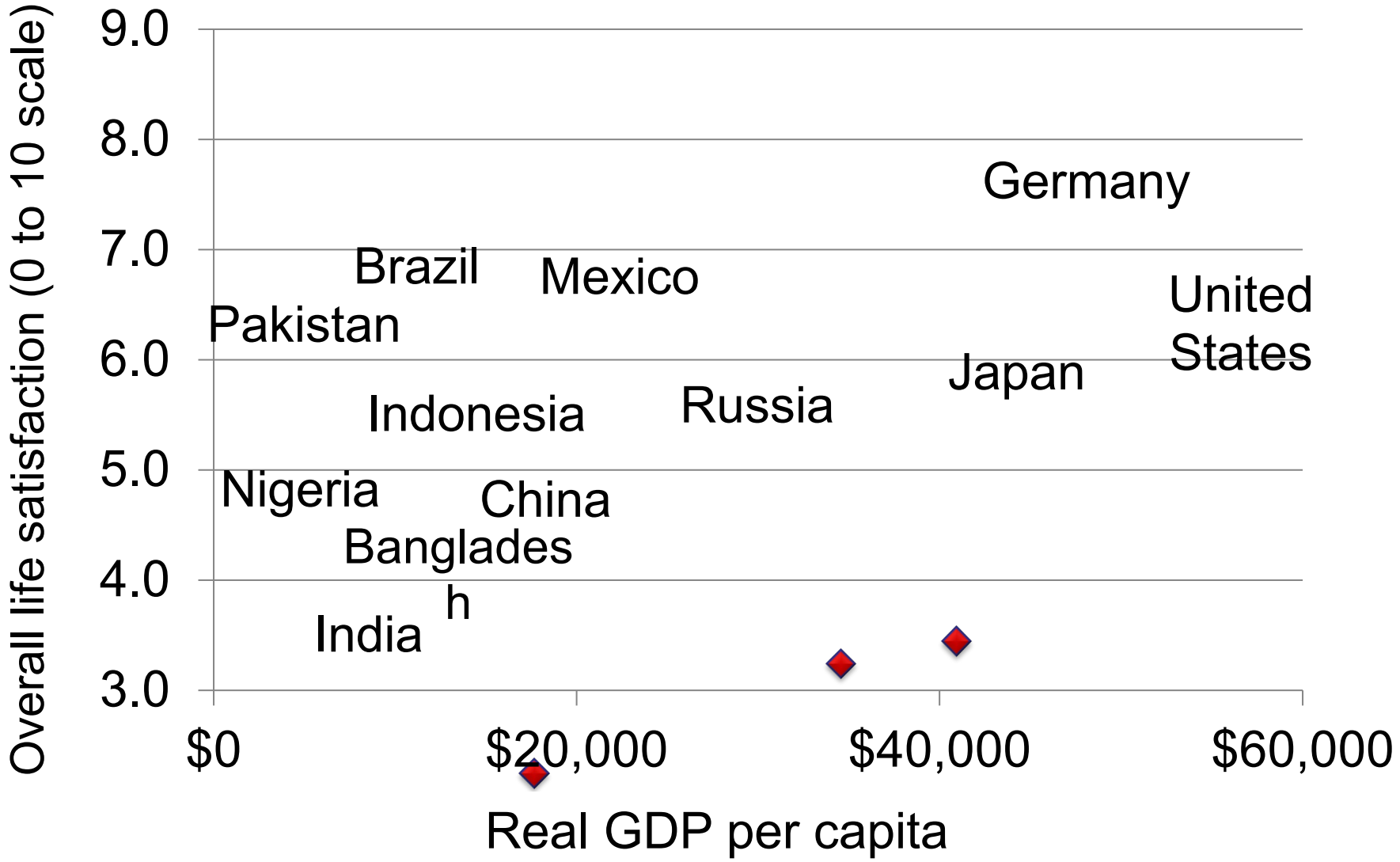
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GDP and Average Schooling in 12 Countries



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GDP and Overall Life Satisfaction in 12 Countries



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CHAPTER IN A NUTSHELL

- Because every transaction has a buyer and a seller, the economy's total expenditure must equal its total income.
- Gross domestic product (GDP) measures an economy's total expenditure on newly produced goods and services and the total income earned from the production of these goods and services. More precisely, GDP is the market value of all final goods and services produced within a country in a given period.

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CHAPTER IN A NUTSHELL

GDP's four components of expenditure: consumption, investment, government purchases, and net exports.

- Consumption includes spending on goods and services by households, with the exception of purchases of new housing.
- Investment includes spending on business capital, residential capital, and inventories.
- Government purchases include spending on goods and services by local, state, and federal governments.
- Net exports equal the value of goods and services produced domestically and sold abroad (exports) minus the value of goods and services produced broad and sold domestically (imports).

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CHAPTER IN A NUTSHELL

- Nominal GDP uses current prices to value the economy's production of goods and services. Real GDP uses constant base-year prices to value this production. The GDP deflator—calculated from the ratio of nominal GDP to real GDP—measures the level of prices in the economy.
- GDP is a good measure of economic well-being because people usually prefer higher to lower incomes. But it is not a perfect measure of well-being. For example, GDP excludes the value of leisure and the value of a clean environment.

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